Chapter 11 exercises

DS11-E01 • B-tree insertion and deletion

Use minimum degree t=2 and top-down splitting. From empty, insert 10,20,5,6,12,30,25, then delete 12, preferring the left sibling when borrowing. Show each split, promotion, borrowed key, and final tree. Check every occupancy requirement.

DS11-E02 • Height and I/O

Derive $n \geq 2t^h - 1$ for a nonempty B-tree with height h≥1. For t=64, n=1,000,000, find the greatest integer height permitted by this bound. Does that inequality exclude height 2? Use maximum capacity as a separate check.

DS11-E03 • B+ separators and deletion

With maximum leaf capacity L=3, insert 10 into `[4,9,12]` and split into two records per leaf. Use right-minimum separators. Draw the root, leaf chain, and record locations. In a separate legal L=2 state with root `[10]` and leaves `[4,9]`, `[10,12]`, delete 10. Which keys and links change?

DS11-E04 • Range-query assumptions

B+ leaves hold keys and record pointers. Integer capacities M≥3 and L≥2 bound internal child counts and leaf record counts. Their nonroot minima are a=ceil(M/2) and ell=ceil(L/2). An internal root has at least two children. Height h counts root-to-leaf edges. Derive general height and index I/Os for k outputs, then state conditions yielding $O(1 + \log_B n + k/B)$. Separate scattered/clustered full records and output work. Construct the M=3,$L = B = 2^q$ full-binary counterexample. Explain why the physical location of child.minimum and persistent parent fields affects update I/Os.

DS11-E05 • Incompatible order conventions

Text A defines order m as maximum children. Text B defines minimum degree t as at least t and at most 2t nonroot children. With m=5, can t=m/2 be substituted into this chapter's split procedure? State capacity and rounding differences.

DS11-E06 • B+ underflow, exact separators, and a range

Let M=3,L=3. Root separators are [10,14], with leaves [4,9]↔[10,12]↔[14,16]. Delete 10,14,4 in order. Show each borrow or merge, parent separators, minimum occupancy, and leaf links. Prefer merging with the left sibling at the first step. After deleting 14, execute inclusive range [9,14], distinguishing inspected keys from output. Explain final root contraction and whether an intervening B-tree parent key could also be discarded.

DS11-E07 • Internal underflow, child transfer, and root contraction

> M=4,L=3; separators are exact right minima. Internal entries are (child_id,cached_min). Root R has P,Q. P owns leaves A=[1,3], B=[5,7]; Q owns C=[9,11], D=[13,15], E=[17,19]. The chain is A↔B↔C↔D↔E and edge-height is two. Delete 1. Try a same-parent sibling before merging; retain the left page on merge.

1. Trace the leaf merge and internal borrowing. Give P.entries,Q.entries,root entries, all separators, and the leaf chain. Trace a search for 9.
2. Independently start without E and delete 1. Give the internal merge and contracted root. What does closed range [7,14] output?
3. A proposed merge also inserts the old root separator as a data record into a leaf. Which key would be duplicated? Check the correct child and separator counts.

DS11-E08 • B-tree capacities and split timing

Compare the maximum child count and minimum non-root key count for order m=3 and minimum degree t=2. Why can an order-3 full two-key node not be split by the t=2 procedure before insertion? Show how it splits after a third key arrives, and when splitting may continue upward.

DS11-E09 • Consecutive 2–3 tree insertions

The root is [40]. Its left internal child [20] has leaves [5,8], [35]; the right internal child [50,60] has leaves [45], [55], [70,80]. Insert 43, 2 and 90 in order. Draw each completed tree and the temporary overflow after inserting 90, list all promoted keys, and check the final key set and leaf depths.

DS11-E10 • Page reads and writes during a split

In an order-3 tree the root is [60], its left internal child [10] has leaves [5,8], [20], and its right internal child [80] has leaves [70], [90]. Insert 2, splitting after insertion. Assume one page per node, an initially empty buffer, search-path pages retained in the buffer, the left split page reusing the original page, and a newly allocated right page. Write each modified page once at the end; a new page needs no read. Exclude allocation, logging and background I/O. List the pages read and written and count each operation.

DS11-E11 • Height bounds for general order

For a nonempty order-m B-tree with $m \geq 3$, let $q = \lceil m/2 \rceil$ and count height h in edges. Derive the minimum and maximum numbers of nodes and keys. Evaluate all four quantities for $m=3$ and $h=2$, and handle the root-leaf case $h=0$.

References and attribution

Lin Cing, handwritten data structures notes. Pat Morin, Open Data Structures, § 14.2. PostgreSQL 18, B-tree implementation.

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https://github.com/xinhi6716/computer-science-self-study